CIVE475
Life Cycle Assessment

Summer 2025

Dr. Alex Milovanoff

Objectives of the course

- **Understand**: Complexity of sustainability issues related to engineering activities and importance of systems perspective

- **Learn method**: Life cycle assessment, as defined by ISO standards, for evaluating products and services

- **Develop solutions**: Apply LCA and software to examine environmental impacts of real-world products/projects

Objectives of the course

- **Reflect**: Be aware of strengths and limitations of LCA
- **Demonstrate**: Ability to independently summarize, analyze, synthesize, and evaluate information on product and services from a wide variety of sources
- **Communicate**: Effectively present insights

About Alexandre

About Darren

About you!

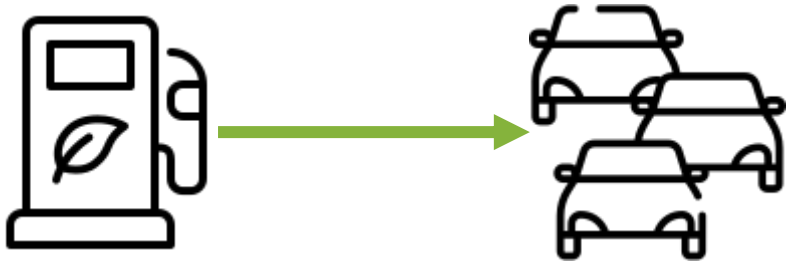
Why should we assess the environmental impacts of our activities?

Is assessing enough?

- Governments seek effective ways to enhance both **economic growth** and **environmental quality**

- **But** public policy issues often framed in terms of **conflict between development** and the **environment**
  - **Does development necessarily require a high environmental price?**
  - **If we must tradeoff between development and the environment, how to decide a proper balance?**

Does development necessarily have a high environmental price?



Issue: Harm caused by growing corn (air pollution and carbon emissions) is equal to about 70% of value of corn produced[1].

Every activity has an environmental impact and cost

But some activities have lower impacts and costs

1: Hill et al. Nature Sustainability. (2019)

What is the goal?

Create an economy with a high level of sustainability through efficient use of resources and prevention of pollution

Why Life Cycle Assessment (LCA)?

- It is an internationally recognized tool to assess the **environmental** efficiency of human activities
- Two main axioms:

“We can’t manage what we don’t measure”

“We can drive business success with environmental stewardship”

Administrative Issues

- **Syllabus** – please review carefully
- **Lectures and readings on Brightspace**
- **Grading scheme**

In-class Exam	20%
Assignment	20%
Project Proposal	10%
Project Literature Review	10%
Project LCA Model	10%
Project Presentation	15%
Project Report	15%

Course Schedule

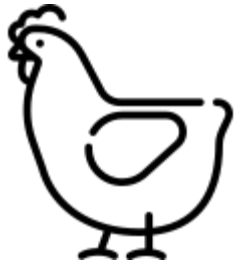
Week, Date of Lecture	Topic of Lecture	Assignment Due (Due Date)
1, May 7	Introductions	-
2, May 14	LCA Framework	-
3, May 21	Goal & Scope Definition	-
4, May 28	Life Cycle Inventory	Project Proposal (May 30)
5, June 4	Multifunctional Processes	-
6, June 11	Life Cycle Impact Assessment	-
7, June 18	ecoinvent	Assignment (June 20)
8, June 25	Uncertainty and Interpretation	Project Literature Review (June 27)
9, July 2	In-class Exam	-
10, July 9	PCR and EPD	Project LCA Model (July 11)
11, July 16	LCA Case Study	-
12, July 23	**NO CLASS**	-
13, July 30	Project Presentation	Project Report (August 8)

Term project

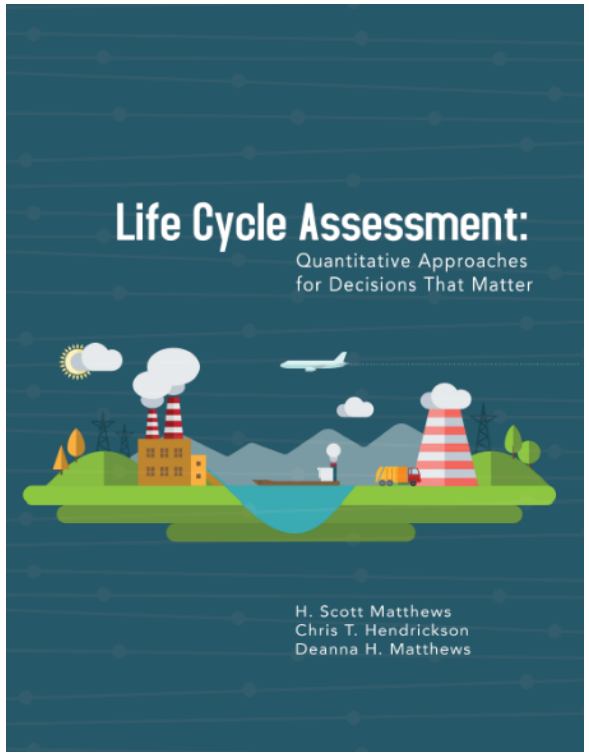
- **Objective:** Conduct a comparative LCA
- Propose a topic that creates synergy with your research, job or interests and where data are available
- Find group members (group max of 4) with common interests

Past course projects - examples

Alternative building materials vs. conventional materials
Reusable vs disposable diapers
Wooden vs. plastic chairs
Water distribution systems
Bridge design materials
Biofuel vehicles vs. electric vehicles in Brazil
Bio-based vs. petroleum-based adipic acid production
Books vs. E-book
Organic eggs vs. conventional effs
Electric bus vs. conventional bus
Straw bale houses vs. conventional houses
Chicken vs. beef
Bottled vs. tap water
Artificial turf vs. natural grass
Regular diet vs. vegetarian diet



Textbook



H. Scott Matthews, Chris T. Hendrickson, Deanna H. Matthews

https://www.lcatextbook.com/

LCA software for tutorials and project

openLca

Questions?

In-Class Activity

1. Find a relevant question that could be answered by a comparative LCA?
   - E.g., "Are electric vehicles better for the environment than conventional vehicles?"
2. Suggest 3 key parameters that could influence the answer to that question
   - E.g., "Emissions intensity of electricity production, weight of the vehicle, purpose of the vehicle"